ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

ENGINEERING MATHEMATICS II

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 2nd year Engineering Mathematics II paper 1 · practice a.

Written by Academic Central for practice, following second-year engineering mathematics modules at South African universities (e.g. University of Pretoria WTW 256 Differential Equations, WTW 258 Calculus and WTW 238 Mathematics). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method and equivalent forms of the answer.

QUESTION 1

1.1 (5)

$\int y^{-2}\,dy = \int x\,dx$ ✓
$-1/y = x^2/2 + C$ ✓
$y(0) = 1$ gives $C = -1$ ✓
$1/y = 1 - x^2/2$ ✓
$y = 2 / (2 - x^2)$ ✓

1.2 (5)

Integrating factor $e^{\int 2\,dx} = e^{2x}$ ✓
$d/dx\,(y\,e^{2x}) = 4e^{2x}$ ✓
$y\,e^{2x} = 2e^{2x} + C$ ✓
$y = 2 + Ce^{-2x}$; $y(0) = 0$ gives $C = -2$ ✓
$y = 2(1 - e^{-2x})$ ✓

1.3 (5)

$T = 20 + 70e^{-kt}$ ✓
$60 = 20 + 70e^{-10k}$, so $e^{-10k} = 4/7$ ✓
$k = \ln(7/4) / 10 = 0.056$ per minute ✓
$30 = 20 + 70e^{-kt}$ gives $e^{-kt} = 1/7$ ✓
$t = \ln 7 / 0.05596 = 34.77$ minutes ✓

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QUESTION 2

2.1 (3)

$r^2 - 5r + 6 = 0$, $r = 2$ or $r = 3$ ✓✓
$y = Ae^{2x} + Be^{3x}$ ✓

2.2 (4)

$r^2 + 4r + 4 = 0$, $r = -2$ (double root) ✓
$y = (A + Bx)e^{-2x}$ ✓
$y(0) = 1$ gives $A = 1$; $y'(0) = B - 2A = 0$ gives $B = 2$ ✓
$y = (1 + 2x)e^{-2x}$ ✓

2.3 (4)

$r^2 + 2r + 5 = 0$, $r = -1 \pm 2j$ ✓✓
$y = e^{-t}(A\cos 2t + B\sin 2t)$ ✓
Complex roots, so the system is underdamped ✓

2.4 (5)

Complementary: $r^2 - 1 = 0$, $y_c = Ae^{x} + Be^{-x}$ ✓
Try $y_p = Ce^{2x}$: $4Ce^{2x} - Ce^{2x} = 3e^{2x}$ ✓
$C = 1$ ✓
$y = Ae^{x} + Be^{-x} + e^{2x}$ ✓✓

2.5 (4)

$y_p = x(C\cos 3x + D\sin 3x)$ ✓✓
$\sin 3x$ and $\cos 3x$ already solve the homogeneous equation ($r = \pm 3j$), so the trial form is multiplied by x ✓
The forcing frequency equals the natural frequency: resonance, and the amplitude grows with time ✓

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QUESTION 3

3.1 $L\{e^{at}\} = \int_0^\infty e^{-st} e^{at}\,dt = \int_0^\infty e^{-(s-a)t}\,dt$ ✓ (3)
$= [-e^{-(s-a)t} / (s-a)]_0^\infty$ ✓
$= 0 + 1/(s-a) = 1/(s-a)$ for $s > a$ ✓

3.2 $6/s^3$ ✓ $- 8/(s^2 + 16)$ ✓ $+ 1/(s+1)$ ✓ (3)

3.3 $(s+5)/((s+1)(s+2)) = A/(s+1) + B/(s+2)$ ✓ (4)
$A = 4$ ✓ $B = -3$ ✓
$f(t) = 4e^{-t} - 3e^{-2t}$ ✓

3.4 $sY - 1 + 2Y = 4/s$ ✓ (5)
$Y = (s+4) / (s(s+2))$ ✓
$= 2/s - 1/(s+2)$ ✓✓
$y = 2 - e^{-2t}$ ✓

3.5 $s^2Y - 1 + 3sY + 2Y = 0$ ✓✓ (5)
$Y = 1 / ((s+1)(s+2)) = 1/(s+1) - 1/(s+2)$ ✓✓
$y = e^{-t} - e^{-2t}$ ✓

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QUESTION 4

4.1 $\partial f/\partial x = 2xy + 3y^2$ ✓ $\partial f/\partial y = x^2 + 6xy$ ✓ $\partial^2 f/\partial x \partial y = 2x + 6y$ ✓ (3)

4.2 $\nabla f(1; 2) = (16; 13)$ ✓ (4)
Unit vector $u = (3/5; 4/5)$ ✓
$D_u f = (48 + 52)/5 = 20$ ✓
Largest rate $= |\nabla f| = \sqrt{(256 + 169)} = 20.62$ ✓

4.3 $f_x = 3x^2 - 3 = 0$ and $f_y = 2y = 0$, so $(1; 0)$ and $(-1; 0)$ ✓✓ (5)
$D = f_{xx} f_{yy} - f_{xy}^2 = (6x)(2) - 0 = 12x$ ✓
$(1; 0)$: $D = 12 > 0$ and $f_{xx} = 6 > 0$, a local minimum, $f = -2$ ✓
$(-1; 0)$: $D = -12 < 0$, a saddle point ✓

4.4 $\int_0^1 [xy + y^2]_0^2\,dx = \int_0^1 (2x + 4)\,dx$ ✓✓ $= 1 + 4 = 5$ ✓ (3)

4.5 $\int_0^{2\pi} \int_0^1 e^{-r^2}\, r\, dr\, d\theta$ ✓✓ (4)
$= 2\pi [-\frac{1}{2} e^{-r^2}]_0^1$ ✓
$= \pi(1 - e^{-1}) = 1.99$ ✓

4.6 $\partial(x)/\partial x = 1 = \partial(y)/\partial y$, so F is conservative, with potential $\varphi = xy$ ✓ (3)
Line integral $= \varphi(1; 1) - \varphi(0; 0)$ ✓ $= 1$ ✓

4.7 $\oint (P\,dx + Q\,dy) = \iint (\partial Q/\partial x - \partial P/\partial y)\,dA = \iint (1 + 1)\,dA$ ✓✓ (3)
$= 2 \times$ area of the circle $= 2\pi$ ✓

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QUESTION 5

5.1 $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0$ ✓ (6)
$\lambda = 5$ and $\lambda = 2$ ✓✓
$\lambda = 5$: $-x_1 + x_2 = 0$, eigenvector $(1; 1)$ ✓
$\lambda = 2$: $2x_1 + x_2 = 0$, eigenvector $(1; -2)$ ✓✓

5.2 $x = c_1e^{5t}(1; 1) + c_2e^{2t}(1; -2)$ ✓✓ (2)

5.3 $a(n+1)/a(n) = (n + 1)/(3n)$ ✓ $\rightarrow 1/3$ as $n \rightarrow \infty$ ✓ (3)
$1/3 < 1$, so the series converges ✓

5.4 f is odd, so $a_0 = a_n = 0$ ✓ (6)
$b_n = (2/\pi) \int$ from 0 to π of $\sin nx\, dx$ ✓ $= (2/(n\pi))(1 - \cos n\pi)$ ✓
$b_n = 4/(n\pi)$ for odd n and 0 for even n ✓
$f(x) = (4/\pi)(\sin x + (\sin 3x)/3 + (\sin 5x)/5 + \ldots)$ ✓✓

5.5 $|a(n+1)/a(n)| = |x|\, n/(n + 1)$ ✓ $\rightarrow |x|$ ✓ (3)
Converges for $|x| < 1$, so $R = 1$ ✓

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TOTAL: 100

END OF MARKING GUIDELINE